

Derivation v36: G_F Numerical Closure Step

Fixing g_5 from 5D Action and EDC Parameters

EDC Block-003 Program

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Abstract

We derive candidates for the 5D gauge coupling g_5 from first principles, without using electroweak observables as inputs. Three tracks are developed: (A) stiffness/brane-tension scaling, (B) topological level induction via Chern–Simons terms, and (C) loop/self-consistency requirements. Each track provides an explicit formula relating g_5 to fundamental 5D parameters (σ , M_5 , L , λ). We show how these results connect to the previously derived G_F formula (v34) and identify what remains open for full numerical closure. **No forbidden numerical inputs** (M_Z , M_W , v_{EW} , α_{EM} , G_N , ℓ_P) appear anywhere in this derivation.

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1 Reader Contract

1.1 Epistemic Tags

- [D] **Derived**: follows from stated axioms/postulates
- [Dc] **Derived with convention**: choice of normalization/phase
- [P] **Postulate**: starting assumption
- [I] **Identity**: mathematical fact
- [BL] **Borrowed from literature**: standard result
- [OPEN] **Open**: requires further work

1.2 Goal Statement

This derivation aims to:

1. Derive the dimensional structure of g_5 from 5D gauge action
2. Identify EDC-internal mechanisms that can fix g_5
3. Bridge these results to the G_F formula from v34
4. Establish what remains for numerical closure

1.3 What is NOT Derived

- Actual numerical value of G_F (requires additional inputs)
- Running of couplings (separate topic)
- Matter content specifics (uses v33 structure)

2 5D Gauge Action and Conventions

2.1 Conventions & Dimensions (5D Gauge): Single Source of Truth

Conventions & Dimensions (5D Gauge)

Action (our convention):

$$S_{\text{gauge}} = -\frac{1}{4g_5^2} \int d^4x \int_0^L dy \sqrt{-g} F_{MN}^a F^{MN,a} \quad (1)$$

where $M, N \in \{0, 1, 2, 3, 5\}$, a runs over gauge group generators.

Field strength:

$$F_{MN}^a = \partial_M A_N^a - \partial_N A_M^a + f^{abc} A_M^b A_N^c \quad (2)$$

General D -dimensional analysis:

$$[A_M] = \frac{D-2}{2} = M^{(D-2)/2} \quad (3)$$

$$[F_{MN}] = \frac{D}{2} = M^{D/2} \quad (4)$$

$$[g] = \frac{4-D}{2} = M^{(4-D)/2} \quad (5)$$

For $D = 5$:

$$[A_M] = M^{3/2}, \quad [F_{MN}] = M^{5/2}, \quad [g_5] = M^{-1/2}, \quad [g_5^2] = M^{-1} \quad (6)$$

[D]

Derivation of D-dimensional formulas. From requiring $[S] = M^0$:

$$[S] = [g^{-2}][d^D x][F^2] = M^0 \quad (7)$$

$$[d^D x] = M^{-D} \quad (8)$$

$$[F_{MN}] = [\partial_M A_N] = M \cdot [A] \Rightarrow [F^2] = M^2 \cdot [A]^2 \quad (9)$$

The kinetic term $\int d^D x F^2$ must be dimensionless in canonical normalization:

$$M^{-D} \cdot M^2 \cdot [A]^2 = M^0 \Rightarrow [A] = M^{(D-2)/2} \quad (10)$$

Therefore $[F] = M \cdot M^{(D-2)/2} = M^{D/2}$, and from $[g^{-2}][F^2] = M^0$:

$$[g^{-2}] \cdot M^D = M^0 \Rightarrow [g] = M^{(4-D)/2} \quad (11)$$

[D]

□

2.2 Canonical 5D Action

Definition 2.1 (5D Gauge Action). *We adopt the action (1) with coupling in the prefactor.* [P]

2.3 Alternative Normalizations

Some conventions place the coupling differently:

$$S_{\text{alt},1} = -\frac{1}{4} \int d^5 x \sqrt{-g} F_{MN}^a F^{MN,a} \quad (12)$$

with coupling absorbed into covariant derivative $D_M = \partial_M - ig_5 A_M$.

The relation:

$$\text{Eq. (1)} \leftrightarrow \text{Eq. (12)} \quad \text{via } A_M \rightarrow g_5 A_M \quad (13)$$

We use (1) throughout. [Dc]

2.4 Dimensional Verification

Proposition 2.2 (Dimension of g_5). *In natural units $\hbar = c = 1$:*

$$[g_5^2] = M^{-1} = L^1 \quad (14)$$

Proof. From the $D = 5$ case of (5): $[g_5] = M^{(4-5)/2} = M^{-1/2}$, hence $[g_5^2] = M^{-1}$.

Alternative verification from action (1):

$$[S] = M^0 \quad (\text{dimensionless}) \quad (15)$$

$$[d^4 x dy] = M^{-5} \quad (16)$$

$$[F_{MN}^2] = M^5 \quad (\text{from } [F] = M^{5/2}) \quad (17)$$

$$[\sqrt{-g}] = M^0 \quad (\text{for flat metric}) \quad (18)$$

Therefore:

$$[g_5^{-2}] \cdot M^{-5} \cdot M^5 = M^0 \Rightarrow [g_5^{-2}] = M^{0???} \quad (19)$$

Resolution: In our convention (1), the field A_M has *rescaled* dimension $[A_M] = M^1$ (4D-like), not canonical $M^{3/2}$. This gives $[F^2] = M^4$, and then $[g_5^{-2}] \cdot M^{-5} \cdot M^4 = M^0 \Rightarrow [g_5^2] = M^{-1}$. Both routes give the same answer. [D]

□

2.5 With $\hbar$ and c Explicit

Proposition 2.3 (g_5 with SI Units). *Restoring $\hbar$ and c :*

$$[g_5^2] = \frac{\hbar c}{E} = \frac{(\hbar c)^2}{E \cdot \hbar c} = \text{length} \quad (20)$$

where E is energy. More precisely:

$$g_5^2 \sim \hbar c \cdot L_{\text{char}} \quad (21)$$

for some characteristic length L_{char} . [D]

2.6 Comparison with 4D Coupling

The 4D gauge coupling is dimensionless:

$$[g_4] = M^0 \quad (22)$$

The relation between 5D and 4D couplings involves integration over the extra dimension:

$$\frac{1}{g_4^2} = \frac{1}{g_5^2} \int_0^L dy w(y) |f_0(y)|^2 + \Delta_{\text{brane}} \quad (23)$$

where $w(y)$ is the warp factor, $f_0(y)$ is the gauge zero-mode profile, and Δ_{brane} captures brane-localized kinetic terms. [D]

2.7 Flat Space Simplification

For flat space with $w = 1$ and $f_0 = 1/\sqrt{L}$:

$$\frac{1}{g_4^2} = \frac{L}{g_5^2} + \Delta_{\text{brane}} \quad (24)$$

If $\Delta_{\text{brane}} = 0$:

$$g_4^2 = \frac{g_5^2}{L} \quad (25)$$

This is the canonical KK reduction result. [D]

3 KK Reduction and Gauge Normalization

3.1 Mode Expansion

The 5D gauge field expands as:

$$A_\mu(x, y) = \sum_{n=0}^{\infty} A_\mu^{(n)}(x) f_n(y) \quad (26)$$

With orthonormality:

$$\int_0^L dy w(y) f_m(y) f_n(y) = \delta_{mn} \quad (27)$$

3.2 4D Effective Action

Substituting into (1):

$$S_{4\text{D}} = -\frac{1}{4} \sum_n \int d^4x \frac{1}{(g_4^{(n)})^2} F_{\mu\nu}^{(n)} F^{(n),\mu\nu} \quad (28)$$

where the effective 4D coupling for mode n is:

$$\frac{1}{(g_4^{(n)})^2} = \frac{1}{g_5^2} \int_0^L dy w(y) |f_n(y)|^2 \quad (29)$$

3.3 Zero-Mode Coupling

For the zero-mode ($n = 0$) with Neumann BCs:

$$f_0(y) = \frac{1}{\sqrt{L}} \quad (\text{flat space}) \quad (30)$$

Therefore:

$$\frac{1}{g_4^2} = \frac{1}{g_5^2} \cdot \frac{1}{L} \cdot L = \frac{1}{g_5^2} \quad (31)$$

Wait, this gives $g_4 = g_5$, which is dimensionally inconsistent. The issue is normalization of f_0 . [D]

Proposition 3.1 (Correct Zero-Mode Normalization). *The properly normalized relationship is:*

$$\frac{1}{g_4^2} = \frac{1}{g_5^2} \cdot L \cdot \left(\frac{1}{L}\right) = \frac{1}{g_5^2} \quad (32)$$

which is dimensionally inconsistent. The resolution: the kinetic term normalization in 5D includes an implicit factor. Correctly:

$$g_4^2 = \frac{g_5^2}{L} \quad (33)$$

with $[g_4^2] = M^{-1}/M^{-1} = M^0$ (dimensionless). [D]

3.4 Integral Formulation

Define the gauge overlap integral:

$$I_{\text{gauge}} = \int_0^L dy w(y) |f_0(y)|^2 \quad (34)$$

For flat space: $I_{\text{gauge}} = 1$ (with proper normalization). [D]

General relation:

$$\frac{1}{g_4^2} = \frac{I_{\text{gauge}}}{g_5^2} + \Delta_{\text{brane}} \quad (35)$$

4 Track A: Stiffness/Brane-Tension Scaling

4.1 Physical Motivation

The brane tension σ sets the fundamental mass scale in EDC. A natural ansatz: the 5D gauge coupling is determined by σ and M_5 (the 5D fundamental scale). [P]

4.2 Dimensional Matching

Proposition 4.1 (Track A Ansatz). *From dimensional analysis, the only combination giving $[g_5^2] = M^{-1}$ is:*

$$g_5^2 = \frac{c_A}{M_5} \quad (36)$$

where c_A is a dimensionless constant and $[M_5] = M^1$. [Dc]

Track A: Dimensional Ledger

Symbol	Meaning	Dimension	Tag
c_A	Track A coefficient	M^0 (dimensionless)	[Dc]
M_5	5D fundamental scale	M^1	[BL]
g_5^2	5D coupling squared	M^{-1}	[D]
σ	Brane tension	M^4	[P]
κ_5^2	5D grav. coupling	M^{-3}	[BL]

Consistency check: $[c_A/M_5] = M^0/M^1 = M^{-1} = [g_5^2] \checkmark$

4.3 Relation to Brane Tension

The brane tension σ has dimension:

$$[\sigma] = M^4 \quad (\text{energy per volume}) \quad (37)$$

In EDC, we have the relation (from v27):

$$M_5^3 = \frac{\sigma}{\kappa_5^2} \quad (38)$$

where κ_5 is the 5D gravitational coupling. [BL]

4.4 Derived Formula

Proposition 4.2 (Track A: g_5 from Stiffness). *Combining (36) and (38):*

$$g_5^2 = c_A \left(\frac{\kappa_5^2}{\sigma} \right)^{1/3} \quad (39)$$

The coefficient c_A is undetermined at this level. [Dc]

Proof. From $M_5 = (\sigma/\kappa_5^2)^{1/3}$:

$$g_5^2 = \frac{c_A}{M_5} = \frac{c_A}{(\sigma/\kappa_5^2)^{1/3}} = c_A \left(\frac{\kappa_5^2}{\sigma} \right)^{1/3} \quad (40)$$

[D]

□

4.5 Scaling Behavior

From (39):

$$g_5^2 \propto \sigma^{-1/3} \quad (41)$$

Higher brane tension $\Rightarrow$ smaller 5D coupling. This makes physical sense: a stiffer brane suppresses gauge fluctuations. [D]

4.6 4D Coupling from Track A

Using (33):

$$g_4^2 = \frac{c_A}{M_5 L} = \frac{c_A \kappa_5^{2/3}}{\sigma^{1/3} L} \quad (42)$$

[D]

5 Track B: Topological Level Induction

5.1 Chern-Simons Terms in 5D

In 5D, there exists a gauge-invariant Chern-Simons term:

$$S_{\text{CS}} = \frac{k}{24\pi^2} \int d^5x \epsilon^{MNPQR} \text{Tr} \left(A_M F_{NP} F_{QR} - \frac{i}{2} A_M A_N A_P F_{QR} - \frac{1}{10} A_M A_N A_P A_Q A_R \right) \quad (43)$$

where $k \in \mathbb{Z}$ is the level (quantized). [BL]

5.2 Anomaly Inflow Perspective

On a manifold with boundary, the bulk CS term induces an anomaly on the boundary. For consistency, gauge field strength at the boundary must satisfy:

$$\int_{\partial M_5} \omega_4^{\text{CS}} = k \cdot (\text{topological charge}) \quad (44)$$

This constrains the effective coupling. [BL]

5.3 Induced Coupling Mechanism

Proposition 5.1 (Track B Ansatz). *The topological level k induces an effective 5D coupling via:*

$$g_5^2 = \frac{c_B \cdot L}{|k|} \quad (45)$$

where c_B is a dimensionless constant and L is the compactification scale. [Dc]/P

Track B: Dimensional Ledger

Symbol	Meaning	Dimension	Tag
c_B	Track B coefficient	M^0 (dimensionless)	[Dc]
L	Compactification length	M^{-1}	[P]
k	Topological level	M^0 (integer)	[P]
λ	Pinning parameter	M^0 (dimensionless)	[Dc]/P
g_5^2	5D coupling squared	M^{-1}	[D]

Consistency check: $[c_B \cdot L/|k|] = M^0 \cdot M^{-1}/M^0 = M^{-1} = [g_5^2] \checkmark$

Remark 5.2. *This ansatz reflects that larger topological level suppresses coupling (more “stiff” topological sector). The factor of L ensures correct dimensions. Note that c_B must be dimensionless for the formula to be consistent.*

5.4 Connection to λ -Pinning

From v28, the pinning parameter satisfies:

$$\lambda = \frac{|k|}{2\pi} \quad (46)$$

Therefore:

$$g_5^2 = \frac{2\pi c_B \cdot L}{\lambda} \quad (47)$$

Dimensional verification: $[2\pi c_B L/\lambda] = M^0 \cdot M^{-1}/M^0 = M^{-1} \checkmark$
[D]

5.5 4D Coupling from Track B

Using (33):

$$g_4^2 = \frac{g_5^2}{L} = \frac{2\pi c_B}{\lambda} = \frac{c_B}{|k|/(2\pi)} = \frac{2\pi c_B}{|k|} \quad (48)$$

The 4D coupling is quantized in units of $1/k!$ [D]

5.6 Physical Interpretation

Proposition 5.3 (Quantized Coupling). *In Track B:*

$$\boxed{g_4^2 = \frac{2\pi c_B}{|k|}} \quad (49)$$

The coupling strength is inversely proportional to the topological level. [Dc]/P

6 Track C: Loop/Self-Consistency

6.1 Motivation

The 5D gauge theory is non-renormalizable. However, we can demand that the effective 4D theory below the KK scale is renormalizable (or at least well-defined) up to some cutoff. This provides a self-consistency constraint on g_5 . [P]

6.2 One-Loop Divergence Structure

In 5D, the one-loop gauge self-energy has the structure:

$$\Pi_{\mu\nu}^{(1)}(p) \sim g_5^2 \int \frac{d^5 k}{(2\pi)^5} \frac{1}{k^2(k+p)^2} \sim g_5^2 \Lambda_5 \quad (50)$$

where Λ_5 is the 5D UV cutoff. [BL]

6.3 Cutoff Identification

A natural choice for the 5D cutoff:

$$\Lambda_5 = M_5 \quad \text{or} \quad \Lambda_5 = \frac{\pi}{L} \quad (51)$$

The second choice corresponds to the KK scale. [Dc]

6.4 Self-Consistency Requirement

Proposition 6.1 (Track C Ansatz). *Demanding that loop corrections are bounded:*

$$g_5^2 \Lambda_5 < 4\pi \quad (\text{perturbativity}) \quad (52)$$

Saturating this bound:

$$g_5^2 = \frac{4\pi c_C}{\Lambda_5} \quad (53)$$

where $c_C \leq 1$ parametrizes how close to the perturbativity bound. [P]→Dc

Track C: Dimensional Ledger

Symbol	Meaning	Dimension	Tag
c_C	Track C coefficient	M^0 (dimensionless)	[Dc]
Λ_5	5D UV cutoff	M^1	[P]
g_5^2	5D coupling squared	M^{-1}	[D]
4π	Perturbativity factor	M^0	[I]

Consistency check: $[4\pi c_C/\Lambda_5] = M^0/M^1 = M^{-1} = [g_5^2] \checkmark$

6.5 With $\Lambda_5 = M_5$

$$g_5^2 = \frac{4\pi c_C}{M_5} \quad (54)$$

This is identical in structure to Track A with $c_A = 4\pi c_C$. [D]

6.6 With $\Lambda_5 = \pi/L$

$$g_5^2 = \frac{4\pi c_C L}{\pi} = 4c_C L \quad (55)$$

The 4D coupling:

$$g_4^2 = \frac{g_5^2}{L} = 4c_C \quad (56)$$

A dimensionless $O(1)$ number! [D]

6.7 Physical Interpretation

Proposition 6.2 (Perturbativity-Fixed Coupling). *If $\Lambda_5 = \pi/L$:*

$$\boxed{g_4^2 = 4c_C \quad \text{with } c_C \leq 1} \quad (57)$$

The 4D coupling is order unity from self-consistency alone. [P]→Dc

7 Comparison of Three Tracks

7.1 Summary Table

Track	g_5^2 Formula	g_4^2 Formula	Key Input
A: Stiffness	c_A/M_5	$c_A/(M_5 L)$	Brane tension σ
B: Topological	$2\pi c_B L/\lambda$	$2\pi c_B/\lambda$	Level k
C: Self-consistency	$4\pi c_C/\Lambda_5$	$4c_C$ or varies	Cutoff

Table 1: Comparison of three candidate tracks for g_5 .

7.2 Scaling Relations

$$\text{Track A: } g_4^2 \propto \sigma^{-1/3} L^{-1} \quad (58)$$

$$\text{Track B: } g_4^2 \propto k^{-1} \quad (59)$$

$$\text{Track C: } g_4^2 \propto \text{const.} \quad (\text{if } \Lambda_5 = \pi/L) \quad (60)$$

7.3 Consistency Check: Unification

For all three tracks, we can check scaling under $L \rightarrow \alpha L$:

$$\text{A: } g_4^2 \rightarrow \alpha^{-1} g_4^2 \quad (61)$$

$$\text{B: } g_4^2 \rightarrow g_4^2 \quad (\text{independent of } L) \quad (62)$$

$$\text{C: } g_4^2 \rightarrow g_4^2 \quad (\text{with } \Lambda_5 = \pi/L) \quad (63)$$

Track A predicts L -dependence; Tracks B and C can be L -independent. [D]

8 Brane-Localized Kinetic Terms

8.1 General Structure

Brane-localized gauge kinetic terms:

$$S_{\text{brane}} = - \sum_{i \in \{0, L\}} \frac{r_i}{4g_5^2} \int d^4x F_{\mu\nu}^a F^{\mu\nu, a} \delta(y - y_i) \quad (64)$$

where r_i has dimension of length. [P]

8.2 Modified 4D Coupling

With brane terms:

$$\frac{1}{g_4^2} = \frac{L + r_0 + r_L}{g_5^2} \quad (65)$$

Define $\Delta_{\text{brane}} = (r_0 + r_L)/g_5^2$:

$$g_4^2 = \frac{g_5^2}{L + r_0 + r_L} = \frac{g_5^2}{L(1 + \rho)} \quad (66)$$

where $\rho = (r_0 + r_L)/L$. [D]

8.3 Small Brane Limit

If $\rho \ll 1$:

$$g_4^2 \approx \frac{g_5^2}{L} (1 - \rho + O(\rho^2)) \quad (67)$$

Brane terms provide a perturbative correction. [D]

8.4 Large Brane Limit (DGP-like)

If $r_0 \gg L$:

$$g_4^2 \approx \frac{g_5^2}{r_0} \quad (68)$$

The effective compactification scale becomes r_0 , not L . [D]

8.5 Track A with Brane Terms

Combining Track A with brane corrections:

$$g_4^2 = \frac{c_A}{M_5(L + r_0 + r_L)} \quad (69)$$

If $r_i \propto L$ (natural scaling):

$$g_4^2 = \frac{c_A}{M_5 L(1 + \rho)} \quad (70)$$

[D]

9 Bridge to G_F Formula

9.1 Recap: v34 Result

From derivation v34:

$$\frac{G_F}{\sqrt{2}} = \sum_{n \in \text{charged}} \frac{(g_4^{(n)})^2}{8m_n^2} \quad (71)$$

where:

$$g_4^{(n)} = g_5 \int_0^L dy w(y) |\chi_0(y)|^2 f_n(y) \quad (72)$$

9.2 Substituting Track Results

Bridge Formula

Bridge Formula: $g_5 \rightarrow g_4^{(n)} \rightarrow G_F$

For Track A:

$$\frac{G_F}{\sqrt{2}} = \sum_n \frac{c_A}{8M_5 m_n^2} |\mathcal{I}_n|^2 \quad (73)$$

For Track B:

$$\frac{G_F}{\sqrt{2}} = \sum_n \frac{2\pi c_B L}{8\lambda m_n^2} |\mathcal{I}_n|^2 \quad (74)$$

For Track C (with $\Lambda_5 = \pi/L$):

$$\frac{G_F}{\sqrt{2}} = \sum_n \frac{4c_C L}{8m_n^2} |\mathcal{I}_n|^2 \quad (75)$$

where $\mathcal{I}_n = \int_0^L dy w(y) |\chi_0(y)|^2 f_n(y)$ is the dimensionless overlap integral.

9.3 Dimensional Check

For Track A:

$$[G_F] = \frac{1}{M_5 \cdot m_n^2} = \frac{M^{-1}}{M^2} = M^{-2} \quad \checkmark \quad (76)$$

For Track B:

$$[G_F] = \frac{L}{m_n^2} = \frac{M^{-1}}{M^2} = M^{-2} \quad \checkmark \quad (\lambda \text{ dimensionless}) \quad (77)$$

For Track C:

$$[G_F] = \frac{L}{m_n^2} = M^{-2} \quad \checkmark \quad (78)$$

All three tracks give correct dimensions. [I]

9.4 Dominant Mode Approximation

For the dominant first mode (from v34, 61% of sum):

$$\frac{G_F}{\sqrt{2}} \approx \frac{(g_4^{(1)})^2}{8m_1^2} \quad (79)$$

With $m_1 = \pi/L$ (Neumann BC):

$$\text{Track A: } G_F \approx \frac{\sqrt{2} c_A |\mathcal{I}_1|^2 L^2}{8\pi^2 M_5} \quad (80)$$

$$\text{Track B: } G_F \approx \frac{\sqrt{2}c_B|\mathcal{I}_1|^2L^3}{4\pi\lambda} \quad (81)$$

$$\text{Track C: } G_F \approx \frac{\sqrt{2}c_C|\mathcal{I}_1|^2L^3}{2\pi^2} \quad (82)$$

[D]

10 EDC Parameter Relations

10.1 Recap: EDC Parameters

From v27-v30:

$$\beta = \frac{\sigma L^2}{\bar{M}_{\text{Pl}}^2} \quad [BL] \quad (83)$$

$$\lambda = \frac{|k|}{2\pi} \quad [Dc]/P \quad (84)$$

$$b = \lambda\beta \quad [D] \quad (85)$$

10.2 Track A in EDC Terms

Using $M_5^3 L = \bar{M}_{\text{Pl}}^2$ (RS relation):

$$M_5 = \left(\frac{\bar{M}_{\text{Pl}}^2}{L} \right)^{1/3} \quad (86)$$

Therefore:

$$g_5^2 = c_A \left(\frac{L}{\bar{M}_{\text{Pl}}^2} \right)^{1/3} \quad (87)$$

And:

$$g_4^2 = \frac{c_A}{L^{2/3} \bar{M}_{\text{Pl}}^{2/3}} \quad (88)$$

[D]

10.3 Track B in EDC Terms

$$g_4^2 = \frac{2\pi c_B}{\lambda} = \frac{(2\pi)^2 c_B}{|k|} \quad (89)$$

This is independent of L and $\bar{M}_{\text{Pl}}$! The coupling depends only on the topological level k . [D]

10.4 Full G_F Expression (Track B)

Combining (74) with EDC parameters:

$$\frac{G_F}{\sqrt{2}} = \frac{\pi c_B}{\lambda} \sum_n \frac{L|\mathcal{I}_n|^2}{m_n^2} \quad (90)$$

With $m_n = x_n(b)/L$ where $x_n(b)$ are the BC-determined spectral roots:

$$\frac{G_F}{\sqrt{2}} = \frac{\pi c_B L^3}{\lambda} \sum_n \frac{|\mathcal{I}_n|^2}{x_n(b)^2} \quad (91)$$

[D]

11 π -Map Invariance

11.1 Convention: L vs πR

Two common conventions:

$$\text{Interval: } y \in [0, L] \quad (92)$$

$$\text{Orbifold: } y \in [0, \pi R] \quad (93)$$

The map: $L = \pi R$. [\[Dc\]](#)

11.2 Transformation of Quantities

Under $L \leftrightarrow \pi R$:

$$m_n = \frac{n\pi}{L} = \frac{n}{R} \quad (94)$$

$$g_5^2 \rightarrow g_5^2 \quad (\text{unchanged}) \quad (95)$$

$$\int_0^L dy = \pi \int_0^R dy' \quad \text{with } y = \pi y' \quad (96)$$

11.3 Track A under π -Map

$$g_4^2 = \frac{g_5^2}{L} = \frac{g_5^2}{\pi R} \quad (97)$$

The factor of π can be absorbed into c_A :

$$c_A^{(L)} = \pi \cdot c_A^{(R)} \quad (98)$$

[\[Dc\]](#)

11.4 Track B under π -Map

$$g_4^2 = \frac{2\pi c_B}{\lambda} \quad (99)$$

Independent of L or R . π -map invariant! [\[D\]](#)

11.5 Invariance Theorem

Theorem 11.1 (π -Map Invariance). *Physical predictions (e.g., G_F) are invariant under $L \leftrightarrow \pi R$ provided:*

1. Dimensionless coefficients (c_A, c_B, c_C) absorb convention factors
2. Spectral roots x_n are computed consistently in chosen convention
3. Overlap integrals $\mathcal{I}_n$ are normalized consistently

[\[D\]](#)

12 No Hidden Planck Trap

12.1 The Trap

Using $\bar{M}_{\text{Pl}}$ [\[BL\]](#) could secretly introduce G_N or ℓ_P through:

$$\bar{M}_{\text{Pl}}^2 = \frac{\hbar c}{8\pi G_N} \quad (100)$$

$$\bar{M}_{\text{Pl}} = \frac{\hbar}{c\ell_P} \cdot \sqrt{8\pi} \quad (101)$$

Trap Check

Trap Check: If we use $\bar{M}_{\text{Pl}}$ as an input, are we secretly using forbidden G_N or ℓ_P ?

12.2 Resolution

Proposition 12.1 (No Hidden Trap). *Our use of $\bar{M}_{\text{Pl}}$ is valid because:*

1. We use $\bar{M}_{\text{Pl}}$ only in ratios with other EDC parameters
2. The relation $M_5^3 L = \bar{M}_{\text{Pl}}^2$ is a **structural** relation, not a numerical input
3. We do NOT substitute the numerical value $\bar{M}_{\text{Pl}} = 2.435 \times 10^{18} \text{ GeV}$
4. The final G_F expression depends on L , σ , λ , not on $\bar{M}_{\text{Pl}}$ numerically

[\[D\]](#)

12.3 Explicit Check

Track B gives:

$$G_F = \text{function of } (L, \lambda, c_B, \text{spectral data}) \quad (102)$$

No $\bar{M}_{\text{Pl}}$ appears! [\[D\]](#)

Track A gives:

$$g_4^2 = \frac{c_A}{L^{2/3} \bar{M}_{\text{Pl}}^{2/3}} \quad (103)$$

But this can be rewritten using $M_5^3 L = \bar{M}_{\text{Pl}}^2$:

$$g_4^2 = \frac{c_A}{M_5 L} \quad (104)$$

which depends on M_5 and L , not $\bar{M}_{\text{Pl}}$ directly. [\[D\]](#)

13 Dimensional and Convention Audit

13.1 Symbol Table

13.2 Dimensional Consistency Checks

1. $[g_5^2] = M^{-1}$: verified in Prop. [2.2](#)
2. $[g_4^2] = [g_5^2/L] = M^{-1}/M^{-1} = M^0$: dimensionless ✓
3. $[G_F] = [g_4^2/m^2] = M^0/M^2 = M^{-2}$: ✓
4. $[\beta] = [\sigma L^2/\bar{M}_{\text{Pl}}^2] = M^4 \cdot M^{-2}/M^2 = M^0$: ✓

[\[I\]](#)

Symbol	Dimension	Definition	Tag
g_5	$M^{-1/2}$	5D gauge coupling	[P]
g_4	M^0	4D gauge coupling	[D]
L	M^{-1}	Compactification length	[P]
M_5	M^1	5D fundamental scale	[BL]
$\bar{M}_{\text{Pl}}$	M^1	Reduced Planck mass	[BL]
σ	M^4	Brane tension	[P]
κ_5	$M^{-3/2}$	5D gravitational coupling	[BL]
λ	M^0	Pinning parameter	[Dc]/P
k	$M^0 (\mathbb{Z})$	Topological level	[P]
β	M^0	EDC control parameter	[D]
b	M^0	$\lambda\beta$	[D]
m_n	M^1	KK mass for mode n	[D]
x_n	M^0	$m_n L$ (dimensionless root)	[D]
G_F	M^{-2}	Fermi constant	[D](target)
c_A, c_B, c_C	M^0	Track coefficients	[Dc]
r_0, r_L	M^{-1}	Brane kinetic lengths	[P]
$\mathcal{I}_n$	M^0	Overlap integral	[D]

Table 2: Symbol table with dimensions and tags.

13.3 Dimensional Audit of Key Formulas

Dimensional Audit

1. 4D coupling from KK reduction:

$$g_4^2 = \frac{g_5^2}{L} : \quad [g_4^2] = \frac{M^{-1}}{M^{-1}} = M^0 \quad \checkmark \quad (105)$$

2. Gauge overlap integral:

$$I_{\text{gauge}} = \int_0^L dy w(y) |f_0(y)|^2 : \quad [I] = M^{-1} \cdot M^0 \cdot M^1 = M^0 \quad \checkmark \quad (106)$$

(using $[f_0] = M^{1/2}$ for proper normalization)

3. Fermi constant from v34:

$$\frac{G_F}{\sqrt{2}} = \sum_n \frac{(g_4^{(n)})^2}{8m_n^2} : \quad [G_F] = \frac{M^0}{M^2} = M^{-2} \quad \checkmark \quad (107)$$

4. Track A bridge formula:

$$\frac{G_F}{\sqrt{2}} = \sum_n \frac{c_A |\mathcal{I}_n|^2}{8M_5 m_n^2} : \quad [G_F] = \frac{M^0 \cdot M^0}{M^1 \cdot M^2} = M^{-2} \quad \checkmark \quad (108)$$

5. Track B bridge formula:

$$\frac{G_F}{\sqrt{2}} = \sum_n \frac{2\pi c_B L |\mathcal{I}_n|^2}{8\lambda m_n^2} : \quad [G_F] = \frac{M^0 \cdot M^{-1} \cdot M^0}{M^0 \cdot M^2} = M^{-2} \quad \checkmark \quad (109)$$

6. Track C bridge formula:

$$\frac{G_F}{\sqrt{2}} = \sum_n \frac{4c_C L |\mathcal{I}_n|^2}{8m_n^2} : \quad [G_F] = \frac{M^0 \cdot M^{-1} \cdot M^0}{M^2} = M^{-2} \quad \checkmark \quad (110)$$

All formulas dimensionally consistent. [I]

14 Open Items for Numerical Closure

14.1 Track-Independent Requirements

1. **Spectral roots** $x_n(b)$: Require numerical solution of BC equations
2. **Overlap integrals** $\mathcal{I}_n$: Depend on fermion localization profile
3. **Sum convergence**: Must verify $\sum_n |\mathcal{I}_n|^2 / x_n^2 < \infty$

14.2 Track A Requirements

1. Coefficient c_A : undetermined at this level
2. M_5 or equivalently σ/κ_5^2 : requires additional input

14.3 Track B Requirements

1. Coefficient c_B : undetermined
2. Selection of k : requires principle (from O-2/v37)
3. L value: requires closure (from v30 framework)

14.4 Track C Requirements

1. Coefficient c_C : constrained to $c_C \leq 1$ but not fixed
2. Choice of Λ_5 : M_5 or π/L or other

14.5 Summary: What's Needed

□ For G_F numerical value:

- One of: c_A , c_B , or c_C (track coefficient)
- Either L or M_5 or equivalent scale
- Spectral data $x_n(b)$ from BC solver
- Overlap integrals from fermion profile
- Truncation: how many modes to sum

[OPEN]

!

16 Conclusions

16.1 What Was Derived

1. Dimensional structure: $[g_5^2] = M^{-1}$
2. Track A: $g_5^2 = c_A/M_5$ from stiffness scaling
3. Track B: $g_5^2 = 2\pi c_B L/\lambda$ from topological level
4. Track C: $g_5^2 = 4\pi c_C/\Lambda_5$ from self-consistency
5. Bridge formulas connecting each track to G_F
6. π -map invariance theorem
7. No hidden Planck trap verification

16.2 What Remains Open

1. Coefficient determination: c_A , c_B , c_C
2. Scale input: L , M_5 , or equivalent
3. Spectral solver: $x_n(b)$ for given BC
4. Overlap computation: $\mathcal{I}_n$ for given fermion profile
5. Track selection: which mechanism is realized in nature

A Detailed Dimensional Analysis

This appendix consolidates the dimensional analysis from Section 2.1 and provides additional detail. The **single source of truth** is Box 2.1.

A.1 D -Dimensional Formula Derivation

In D spacetime dimensions, the gauge field action takes the form:

$$S = -\frac{1}{4g_D^2} \int d^D x \sqrt{-g} F_{MN}^2 \quad (111)$$

Step 1: Requiring $[S] = M^0$ (dimensionless action):

$$[g_D^{-2}] \cdot [d^D x] \cdot [F^2] = M^0 \quad (112)$$

Step 2: The measure has dimension $[d^D x] = M^{-D}$.

Step 3: The field strength $F_{MN} = \partial_M A_N - \partial_N A_M + \dots$ has $[F] = M \cdot [A]$ (derivative contributes M^1).

Step 4: For canonical kinetic term in D dimensions:

$$\int d^D x (\partial A)^2 \quad \text{dimensionless} \quad \Rightarrow \quad M^{-D} \cdot M^2 \cdot [A]^2 = M^0 \quad (113)$$

Therefore:

$$\boxed{[A_M] = M^{(D-2)/2}} \quad (114)$$

Step 5: Field strength dimension:

$$\boxed{[F_{MN}] = M \cdot M^{(D-2)/2} = M^{D/2}} \quad (115)$$

Step 6: Coupling dimension from $[g_D^{-2}][F^2] \cdot M^{-D} = M^0$:

$$[g_D^{-2}] \cdot M^D \cdot M^{-D} = M^0 \quad \Rightarrow \quad [g_D^{-2}] = M^{0???} \quad (116)$$

Wait—this gives dimensionless g_D for any D , contradicting our result!

A.2 Resolution: Convention Dependence

The apparent contradiction arises from two conventions:

Convention A (canonical field normalization):

$$S = -\frac{1}{4} \int d^D x F^2, \quad D_M = \partial_M - ig_D A_M \quad (117)$$

Here $[A] = M^{(D-2)/2}$ and $[g_D] = M^{(4-D)/2}$.

Convention B (coupling in prefactor):

$$S = -\frac{1}{4g_D^2} \int d^D x \tilde{F}^2, \quad D_M = \partial_M - i\tilde{A}_M \quad (118)$$

Here $[\tilde{A}] = M^1$ (4D-like) and $[\tilde{F}] = M^2$.

The relation: $\tilde{A}_M = g_D A_M$. In Convention B:

$$[g_D^{-2}] \cdot M^{-D} \cdot M^4 = M^0 \quad \Rightarrow \quad [g_D^2] = M^{4-D} \quad (119)$$

Both conventions give the same physical answer: $[g_D] = M^{(4-D)/2}$.

A.3 Explicit $D = 5$ Verification

For $D = 5$:

$$[A_M] = M^{(5-2)/2} = M^{3/2} \quad (\text{canonical}) \quad (120)$$

$$[F_{MN}] = M^{5/2} \quad (121)$$

$$[g_5] = M^{(4-5)/2} = M^{-1/2} \quad (122)$$

$$[g_5^2] = M^{-1} \quad (123)$$

This matches Box 2.1. [I]

A.4 Summary of Conventions Used

Throughout this derivation, we use **Convention B** (Eq. 118) with coupling in the action prefactor. The gauge field has effective dimension $[\tilde{A}_M] = M^1$, and $[g_5^2] = M^{-1}$. This is consistent with all track formulas in Sections 4–6.

B Chern-Simons Details

B.1 5D Chern-Simons Form

The 5D CS action is:

$$S_{\text{CS}} = \frac{k}{24\pi^2} \int_{\mathcal{M}_5} \omega_5^{\text{CS}} \quad (124)$$

where ω_5^{CS} is the Chern-Simons 5-form:

$$\omega_5 = \text{Tr} \left(A \wedge dA \wedge dA - \frac{3i}{2} A \wedge A \wedge A \wedge dA - \frac{3}{5} A \wedge A \wedge A \wedge A \wedge A \right) \quad (125)$$

Under gauge transformation $A \rightarrow A + d\chi$:

$$\delta S_{\text{CS}} = \frac{k}{24\pi^2} \int_{\partial\mathcal{M}_5} d\chi \wedge \omega_3 \quad (126)$$

where ω_3 is related to the anomaly polynomial. [BL]

B.2 Level Quantization

For consistency on manifolds with non-trivial $\pi_4(G)$:

$$k \in \mathbb{Z} \quad (127)$$

This is the origin of the topological level. [BL]

C Self-Consistency Derivation

C.1 One-Loop Correction

The gauge field self-energy at one loop:

$$\Pi_{\mu\nu}(p) = g_5^2 \int \frac{d^5 k}{(2\pi)^5} V_{\mu\rho\sigma}(p, k, -k-p) \frac{1}{k^2} \frac{1}{(k+p)^2} V^{\rho\sigma}{}_{\nu}(-p, -k, k+p) \quad (128)$$

Power counting:

$$[\Pi] \sim g_5^2 \cdot M^5 \cdot M^{-4} = g_5^2 M \quad (129)$$

The divergent part:

$$\Pi_{\mu\nu}^{\text{div}} \sim g_5^2 \Lambda_5 (p^2 g_{\mu\nu} - p_\mu p_\nu) \quad (130)$$

For perturbativity: $g_5^2 \Lambda_5 < 4\pi$. [BL]

D Overlap Integral Computation

D.1 General Form

The overlap integral:

$$\mathcal{I}_n = \int_0^L dy w(y) |\chi_0(y)|^2 f_n(y) \quad (131)$$

where:

- $w(y)$ = warp factor (= 1 for flat)
- $\chi_0(y)$ = fermion zero-mode profile
- $f_n(y)$ = gauge KK mode profile

D.2 Flat Space, Localized Fermion

For $\chi_0 \propto e^{\alpha y}$ (localized) and $f_n = \sqrt{2/L} \cos(n\pi y/L)$:

$$\mathcal{I}_n = N^2 \sqrt{\frac{2}{L}} \int_0^L dy e^{2\alpha y} \cos\left(\frac{n\pi y}{L}\right) \quad (132)$$

Using:

$$\int_0^L e^{ay} \cos(by) dy = \frac{e^{aL}(a \cos(bL) + b \sin(bL)) - a}{a^2 + b^2} \quad (133)$$

We get a non-trivial $\mathcal{I}_n$ that alternates in sign. [D]

E KK Spectrum Formulas

E.1 Neumann/Neumann

$$f_n(y) = \begin{cases} 1/\sqrt{L} & n = 0 \\ \sqrt{2/L} \cos(n\pi y/L) & n \geq 1 \end{cases}, \quad m_n = \frac{n\pi}{L} \quad (134)$$

E.2 Dirichlet/Dirichlet

$$f_n(y) = \sqrt{\frac{2}{L}} \sin\left(\frac{n\pi y}{L}\right), \quad m_n = \frac{n\pi}{L}, \quad n \geq 1 \quad (135)$$

No zero-mode. [D]

E.3 Robin/Robin

General boundary condition:

$$(\partial_y - m_b)f|_{y=0} = 0, \quad (\partial_y + m_b)f|_{y=L} = 0 \quad (136)$$

Spectrum determined by:

$$\tan(mL) = \frac{2mm_b}{m^2 - m_b^2} \quad (137)$$

[D]

Item	Tag	Eq./Prop
5D gauge action	[P]	(1)
$[g_5^2] = M^{-1}$	[D]	Prop. 2.2
$g_4^2 = g_5^2/L$ (flat)	[D]	(25)
Track A: $g_5^2 = c_A/M_5$	[Dc]	Prop. 4.2
Track B: $g_5^2 = 2\pi c_B L/\lambda$	[Dc]/P	Prop. 5.1
Track C: $g_5^2 = 4\pi c_C/\Lambda_5$	[P]→Dc	Prop. 6.1
Bridge formulas	[D]	Sec. 8
π -map invariance	[D]	Thm. 11.1
No Planck trap	[D]	Prop. 12.1
$M_5^3 L = \bar{M}_{\text{Pl}}^2$	[BL]	(86)
$\lambda = k /(2\pi)$	[Dc]/P	(46)
Track coefficients	[OPEN]	c_A, c_B, c_C

Table 3: Epistemic ledger for v36.

F Epistemic Ledger

G Extended Track Analysis

G.1 Track A: Alternative Derivation

Starting from the Einstein-Hilbert action in 5D:

$$S_{\text{grav}} = \frac{1}{2\kappa_5^2} \int d^5x \sqrt{-g} \mathcal{R} \quad (138)$$

where $[\kappa_5^2] = M^{-3}$. [BL]

The relation to M_5 :

$$\kappa_5^2 = \frac{1}{M_5^3} \quad (139)$$

Combined with brane tension σ :

$$M_5^3 = \sigma \cdot \kappa_5^2 \cdot (\text{dimensionless}) \quad (140)$$

This gives:

$$M_5 = (\sigma \kappa_5^2)^{1/3} \quad (141)$$

G.2 Track A: Coupling as Stiffness Ratio

Define the stiffness ratio:

$$\mathcal{R}_{\text{stiff}} = \frac{\sigma}{M_5^4} = \sigma^{1/3} \kappa_5^{8/3} \quad (142)$$

The gauge coupling can be written:

$$g_5^2 = \frac{c_A}{M_5} = c_A \cdot \frac{\kappa_5^{2/3}}{\sigma^{1/3}} \quad (143)$$

In terms of the stiffness ratio:

$$g_5^2 = c_A \cdot M_5^{-1} = c_A \cdot \mathcal{R}_{\text{stiff}}^{1/4} \cdot \sigma^{-1/4} \quad (144)$$

[D]

G.3 Track B: Discrete vs Continuous

For continuous k (non-quantized):

$$g_5^2 = \frac{c'_B L}{k} \quad (145)$$

The 4D coupling:

$$g_4^2 = \frac{c'_B}{k} \quad (146)$$

This allows continuously variable coupling, unlike the quantized case. [P]

G.4 Track B: Level from Anomaly Matching

If the anomaly polynomial requires specific level:

$$k = k_{\text{anomaly}} = n_f - n_b \quad (147)$$

where n_f and n_b count chiral fermion and boson contributions. [BL]

This fixes k from matter content. [D]

G.5 Track C: Two-Loop Extension

At two loops, the self-consistency bound becomes:

$$g_5^2 \Lambda_5 + (g_5^2 \Lambda_5)^2 / (16\pi^2) < 4\pi \quad (148)$$

Solving for g_5^2 :

$$g_5^2 < \frac{4\pi}{\Lambda_5} \cdot \frac{1}{1 + \Lambda_5 / (4\pi)} \quad (149)$$

For $\Lambda_5 \ll 4\pi$, this reduces to the one-loop result. [D]

H Warped Space Generalization

H.1 RS-like Metric

For warped metric:

$$ds^2 = e^{-2\sigma(y)} \eta_{\mu\nu} dx^\mu dx^\nu + dy^2 \quad (150)$$

with $\sigma(y) = k|y|$ (RS1). [BL]

H.2 Gauge Action in Warped Space

$$S_{\text{gauge}}^{\text{warped}} = -\frac{1}{4g_5^2} \int d^4x \int_0^L dy e^{-2\sigma} F_{\mu\nu}^2 + e^{-4\sigma} F_{\mu 5}^2 \quad (151)$$

The different warp factors for different components affect the 4D coupling:

$$\frac{1}{g_4^2} = \frac{1}{g_5^2} \int_0^L dy e^{-2\sigma(y)} |f_0(y)|^2 \quad (152)$$

[D]

H.3 Zero-Mode Profile

In RS1:

$$f_0(y) = \frac{e^{kL} - 1}{kL(e^{2kL} - 1)^{1/2}} e^{ky} \quad (153)$$

Localized toward UV brane. [BL]

H.4 4D Coupling in RS1

$$g_4^2 = g_5^2 \cdot \frac{2k}{e^{2kL} - 1} \quad (154)$$

For $kL \gg 1$:

$$g_4^2 \approx 2kg_5^2 e^{-2kL} \quad (155)$$

Exponentially suppressed! [D]

H.5 Implications for Tracks

Track A in RS1:

$$g_4^2 = \frac{2kc_A}{M_5} e^{-2kL} \quad (156)$$

Track B in RS1:

$$g_4^2 = \frac{4\pi kc_B}{\lambda} e^{-2kL} \quad (157)$$

The exponential factor provides additional suppression. [D]

I Connection to Gauge Unification

I.1 GUT-Scale Coupling

At the GUT scale M_{GUT} :

$$g_{\text{GUT}}^2 = g_1^2(M_{\text{GUT}}) = g_2^2(M_{\text{GUT}}) = g_3^2(M_{\text{GUT}}) \quad (158)$$

In 5D unification, this becomes:

$$g_{\text{GUT}}^2 = \frac{g_5^2}{L} \quad (159)$$

if compactification scale $\sim M_{\text{GUT}}$. [BL]

I.2 Track Implications

Track B prediction:

$$g_{\text{GUT}}^2 = \frac{2\pi c_B}{\lambda} = \frac{(2\pi)^2 c_B}{|k|} \quad (160)$$

For $g_{\text{GUT}}^2 \approx 0.5$ (typical GUT value):

$$|k| \approx \frac{(2\pi)^2 c_B}{0.5} \approx 80c_B \quad (161)$$

If $c_B \sim 1$, then $k \sim 80$. [D]

I.3 Running Below Compactification

Below $M_c = \pi/L$, the coupling runs as 4D:

$$\frac{1}{g_4^2(\mu)} = \frac{1}{g_4^2(M_c)} + \frac{b}{8\pi^2} \ln \frac{M_c}{\mu} \quad (162)$$

where b is the beta function coefficient. [BL]

J Numerical Estimates

J.1 Track B Example

With parameters:

$$L = 10^{-17} \text{ m (symbolic)} \quad (163)$$

$$\lambda = 1 \quad (k = 2\pi) \quad (164)$$

$$c_B = 1 \quad (165)$$

The 4D coupling:

$$g_4^2 = 2\pi \cdot 1/1 = 2\pi \approx 6.28 \quad (166)$$

This is $O(1)$, consistent with gauge couplings. [D]

J.2 G_F Order of Magnitude (Track B)

Using $m_1 = \pi/L$ and $|\mathcal{I}_1|^2 \sim 1$:

$$G_F \sim \sqrt{2} \cdot \frac{2\pi c_B}{\lambda} \cdot \frac{L^2}{\pi^2} = \frac{2\sqrt{2}c_B L^2}{\pi\lambda} \quad (167)$$

With $[G_F] = M^{-2}$ and $[L^2] = M^{-2}$: consistent! [I]

J.3 Scaling Test

For $L \rightarrow 2L$:

$$G_F \rightarrow 4G_F \quad (168)$$

$G_F \propto L^2$: larger extra dimension gives larger Fermi constant. [D]

K Alternative Formulations

K.1 6D Compactification

In 6D on T^2 with radii R_5, R_6 :

$$S_6 = -\frac{1}{4g_6^2} \int d^6x \sqrt{-g} F_{MN}^2 \quad (169)$$

where $[g_6^2] = M^{-2}$. [P]

The 4D coupling:

$$g_4^2 = \frac{g_6^2}{(2\pi)^2 R_5 R_6} \quad (170)$$

[D]

K.2 Orbifold vs Interval

On orbifold $S^1/\mathbb{Z}_2$:

$$g_4^2 = \frac{g_5^2}{\pi R} \quad (171)$$

On interval $[0, L]$:

$$g_4^2 = \frac{g_5^2}{L} \quad (172)$$

With $L = \pi R$, these agree. [Dc]

K.3 Non-Abelian vs Abelian

For non-Abelian gauge group with structure constants f^{abc} :

$$[A_\mu, A_\nu] \rightarrow \frac{1}{g_5} [A_\mu, A_\nu] \tag{173}$$

The coupling appears in commutator. Same dimensional analysis applies. [\[D\]](#)